

PAPER A



2017 SCIENCE

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

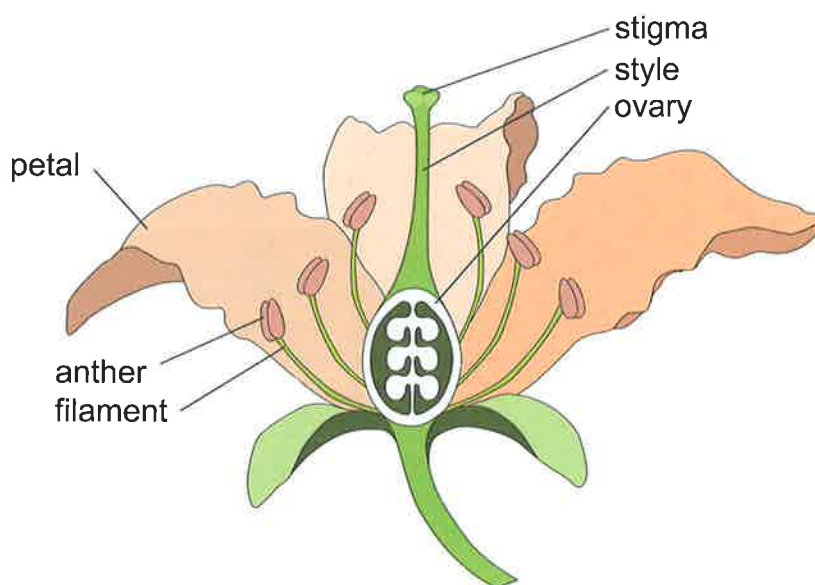
1. Lisa found a frog in her garden. She held her finger near the frog and took this photo.



What does her finger help to show?

- (A) the size of the frog
- (B) the colour of the frog
- (C) the age of the frog
- (D) the shape of the frog

2. This diagram shows some parts of a flower.



On top of which part of the flower is the anther found?

- (A) stigma
- (B) filament
- (C) style
- (D) ovary

3. Paula's teacher gave her four photos, each one showing a different object in space.

photo 1



photo 2



photo 3



photo 4



She also told Paula four facts. Each fact was about one of these objects in space.

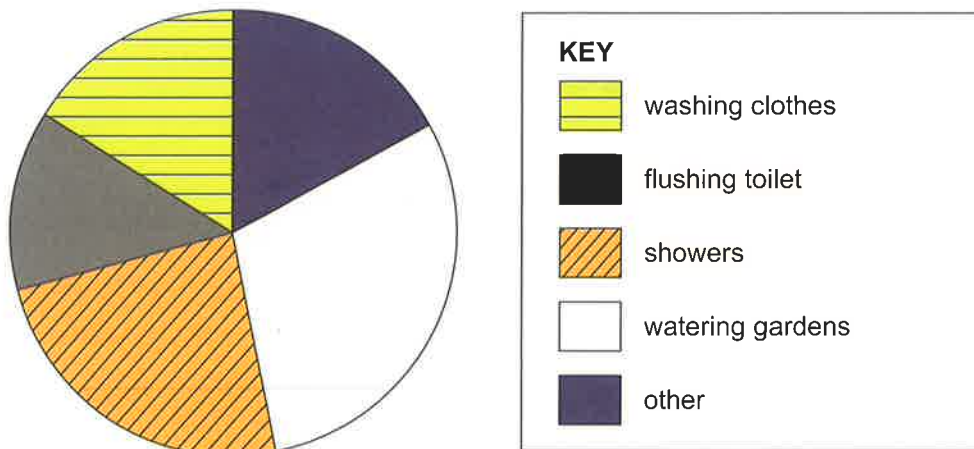
- Saturn is a planet with rings around it.
- A comet is a ball of frozen gases, leaving a trail behind it.
- A meteoroid is a lump of rock travelling in space.
- Mars is a red planet.

Which photo shows a meteoroid?

- (A) photo 1 (B) photo 2 (C) photo 3 (D) photo 4

4. The pie chart compares the amount of water used for different activities in the average home.

Water use in homes



Which activity uses about one quarter of the total water used in the average home?

- (A) washing clothes
(B) showers
(C) watering gardens
(D) other

5. Ruisi works at a panda sanctuary. One of her duties is to check that the newborn pandas are growing at a normal rate.

The table lists how much pandas should weigh at different ages.

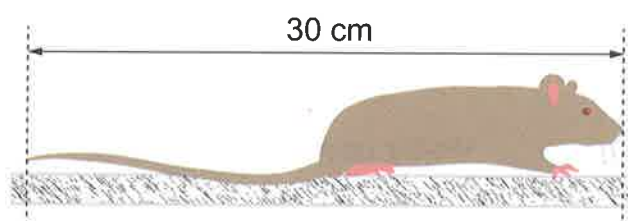
Age	Expected mass (kilograms)
1 week	1
4 weeks	2
12 weeks	5
52 weeks	40

Ruisi is weighing a panda cub that is 6 weeks old.

How much is the panda cub expected to weigh?

- (A) less than 1 kilogram
- (B) between 1 and 2 kilograms
- (C) between 2 and 5 kilograms
- (D) more than 5 kilograms

6. This is a drawing of a rat.



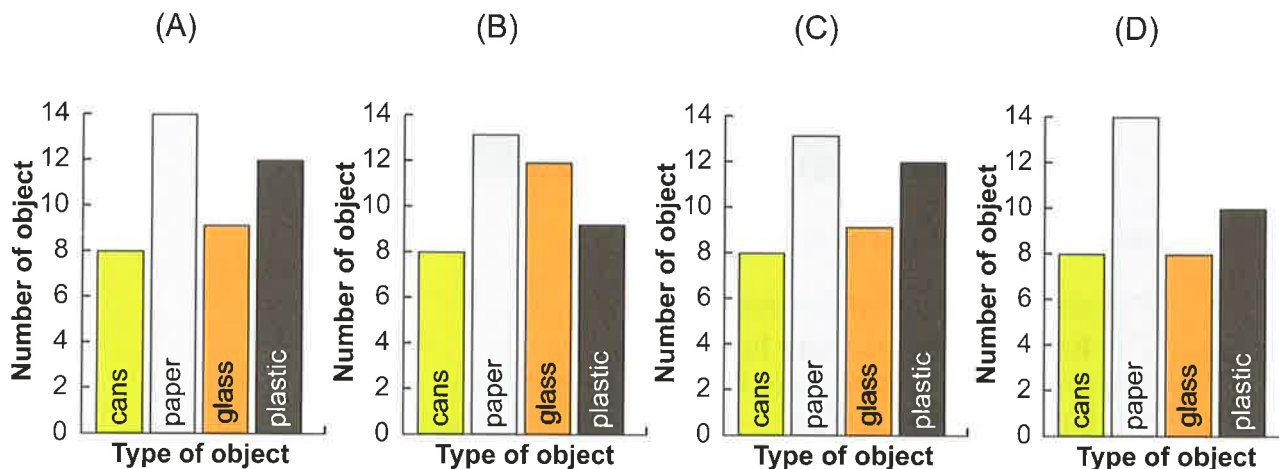
About how many centimetres long is the rat's tail?

- (A) 5
- (B) 10
- (C) 15
- (D) 20

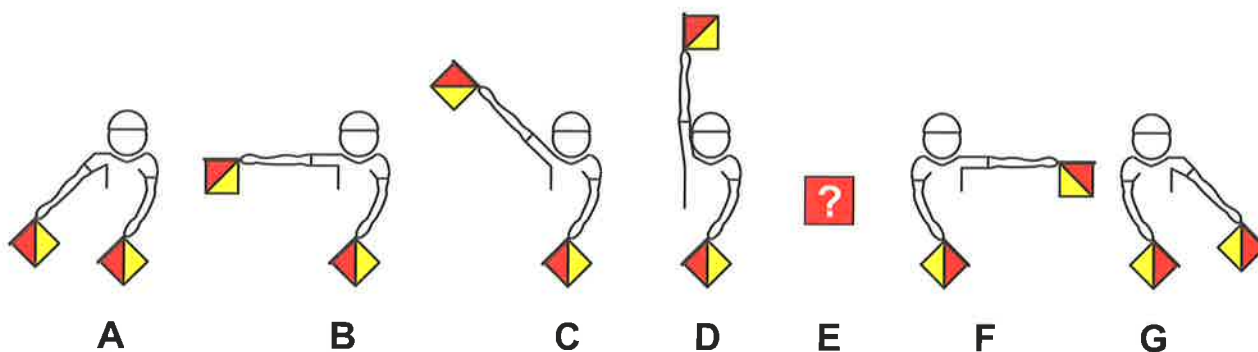
7. Class 3B counted the objects that were collected for recycling in their classroom during one day. They recorded their data in this table.

Type of object being recycled	Number
aluminium cans	8
sheets of paper	14
glass bottles	9
plastic bottles	12

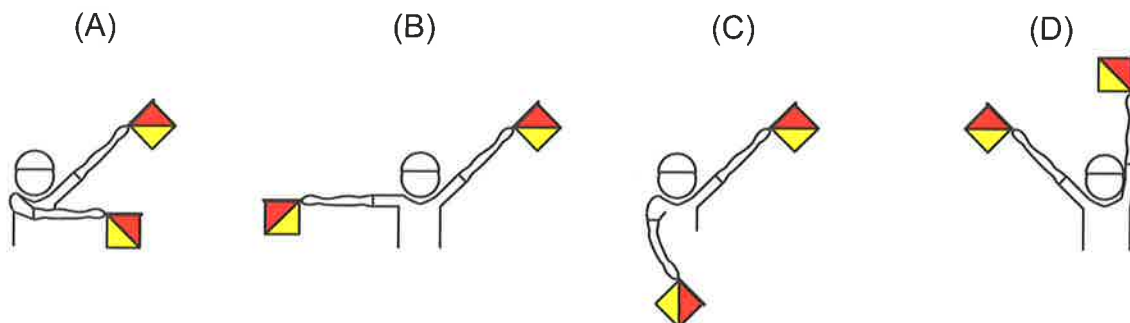
Which graph matches Class 3B's data?



8. Flags can be used to send messages. The position of a pair of flags held by a person matches a letter of the alphabet. The positions of the flags follow a pattern.



Which of the following represents the letter E?



9. Joe has a tree in his paddock.



What causes the tree to have a shadow?

- (A) The shadow is part of the tree.
- (B) The top of the tree is close to the sun.
- (C) The tree needs a shadow to keep it cool in the sun.
- (D) The tree stops the sunlight from reaching the ground.

For questions 10 and 11 use the information below.

The table shows the fastest speed that some animals can run.

Animal	Fastest speed (km/h)
cheetah	120
emu	48
giraffe	51
greyhound dog	63
human	44
polar bear	30

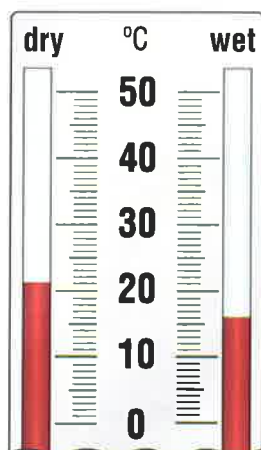
10. Which animals can run faster than a giraffe?

- (A) emu and cheetah
- (B) greyhound dog and human
- (C) cheetah and greyhound dog
- (D) human and emu

11. How much faster than a human can a cheetah run?

- (A) 120 km/h
- (B) 86 km/h
- (C) 76 km/h
- (D) 44 km/h

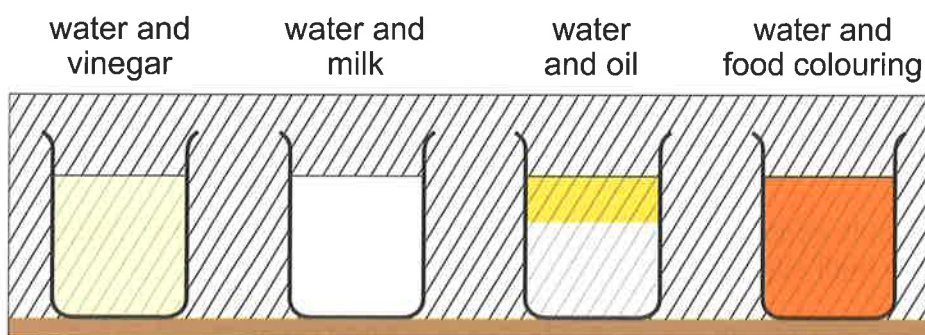
12. A wet and dry bulb thermometer contains two thermometers. The difference between two temperatures on the thermometers can be used to calculate the amount of moisture in the air.



What is the difference between the two temperatures shown on this thermometer?

- (A) 5°C (B) 6°C (C) 10°C (D) 16°C
13. A solution is made when two liquids mix and become a clear liquid. An emulsion is made when two liquids mix but become a cloudy liquid. Some liquids do not mix and form separate layers.

Jack mixed some liquids with water and placed them in front of a patterned board.



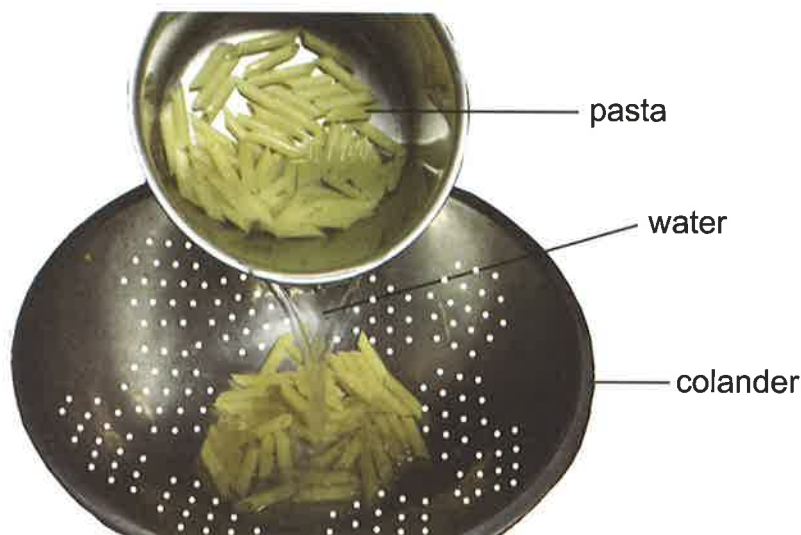
Which of these makes an emulsion when mixed with water?

- (A) vinegar
(B) milk
(C) oil
(D) food colouring

14. Filters can be used to separate solids from liquids.

The liquid that passes through the filter is called the filtrate. The solid that is left behind is called the residue.

Michelle cooked some pasta in water. She then drained the water using a colander as shown.



Which option correctly matches each item with its scientific term?

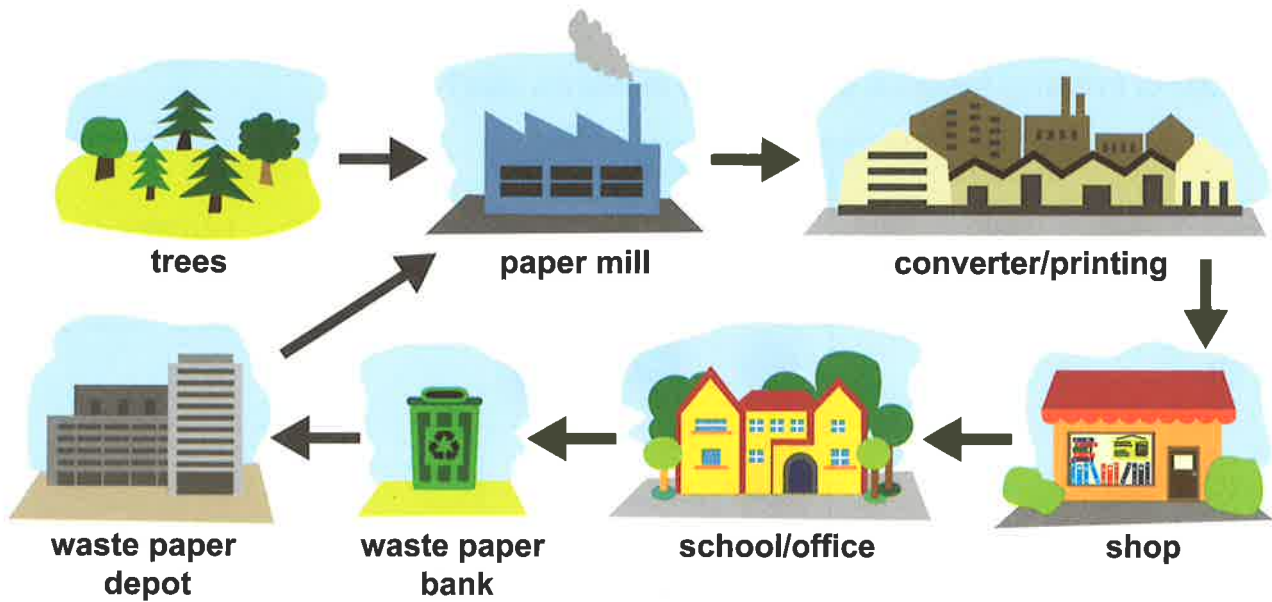
	Pasta	Hot water	Colander
(A)	filtrate	residue	filter
(B)	residue	filter	filtrate
(C)	filter	filtrate	residue
(D)	residue	filtrate	filter

15. Ben lives in South Africa. Ben's dad tells him that from January the daylight hours will get less every day until midyear.

What does Ben need to do every day to find out if this is true?

- (A) Record the time he goes to bed and the time he wakes up.
- (B) Record the time when the Sun is right overhead.
- (C) Record the morning and evening temperatures.
- (D) Record the time of sunrise and sunset.

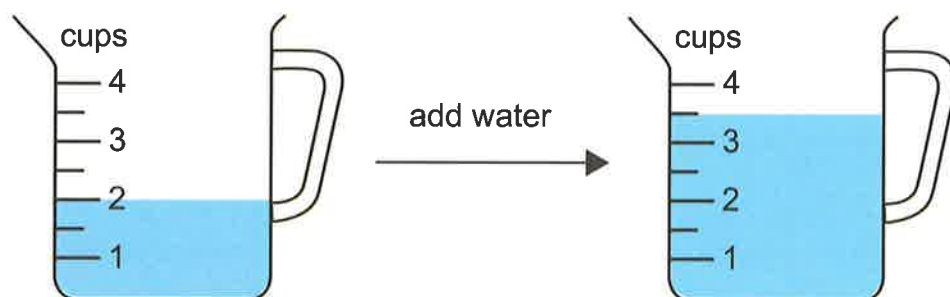
16. The diagram shows the steps for making, using and recycling paper in everyday life.



According to the diagram, what are the steps for the used paper from schools to be recycled to the paper mill?

- (A) → waste paper bank → paper mill
- (B) → waste paper depot → paper mill → trees
- (C) → waste paper bank → waste paper depot → paper mill
- (D) → waste paper depot → waste paper bank → trees

17. Lu had a jug containing some water. She then added some more water to it.



How many cups of water did Lu add?

- (A) 1
- (B) $1\frac{1}{2}$
- (C) 2
- (D) $3\frac{1}{2}$

18. Plants need sunlight to grow.

Paul thinks sunflowers will grow taller outdoors than indoors as they will get more sunlight.

Which of following setups is the best way to test his idea?

	Indoor			Outdoor		
(A)						
(B)						
(C)						
(D)						

KEY

 1 sunflower plant

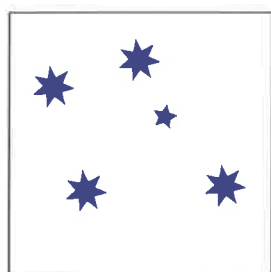
19. The table shows the number of days it takes for different seeds to germinate when planted in soil at different temperatures.

Days for seeds to germinate at different temperatures				
Seeds	10 °C	15 °C	20 °C	25 °C
onion	13	7	5	4
spinach	12	7	6	5
lettuce	7	4	3	2
cabbage	17	10	7	6
carrot	17	10	7	6
pea	14	9	8	6
tomato	43	14	8	6
corn	22	12	7	4
bean	–	30	17	6
cucumber	–	13	6	4

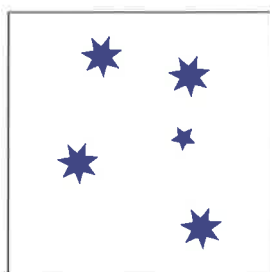
Which is the fastest germinating seed at any temperature?

- (A) spinach (B) tomato (C) cucumber (D) lettuce

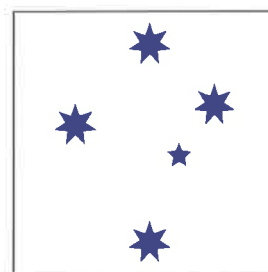
20. An astronomer observed a group of stars called the Southern Cross in the night sky. He drew the position of the stars every two hours, as shown in the diagram.



6 pm



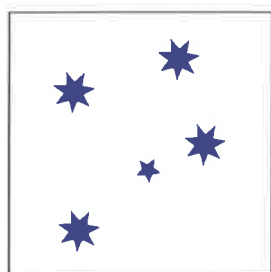
8 pm



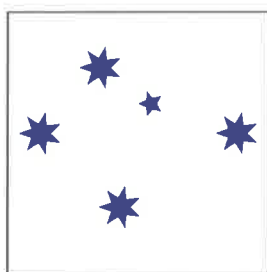
10 pm

Which of these shows the position of the Southern Cross at midnight?

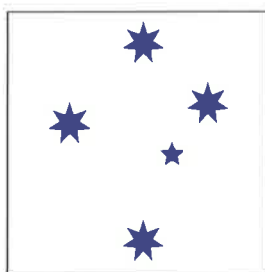
(A)



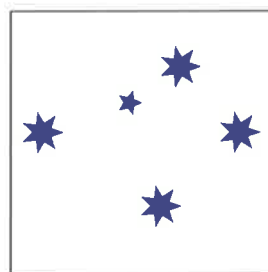
(B)



(C)



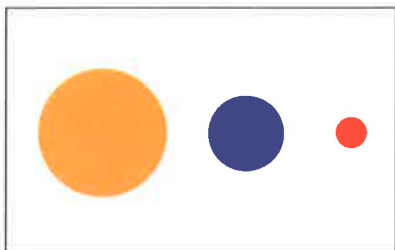
(D)



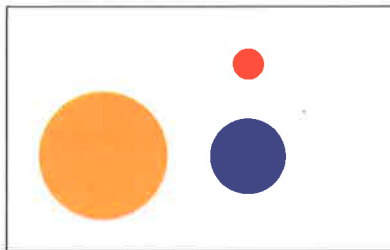
21. A solar eclipse happens when the Moon fully or partially blocks our view of the Sun.

Which diagram shows the positions of the Sun, Earth and Moon when a solar eclipse happens?

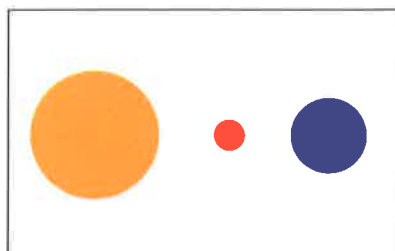
(A)



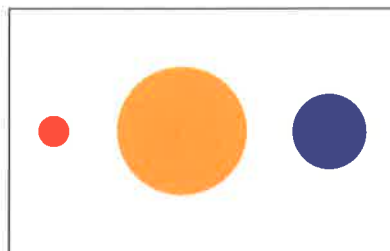
(B)



(C)



(D)



KEY



Sun



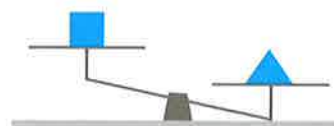
Earth



Moon

NOT TO SCALE

22. Juno used a balance to weigh some blocks.

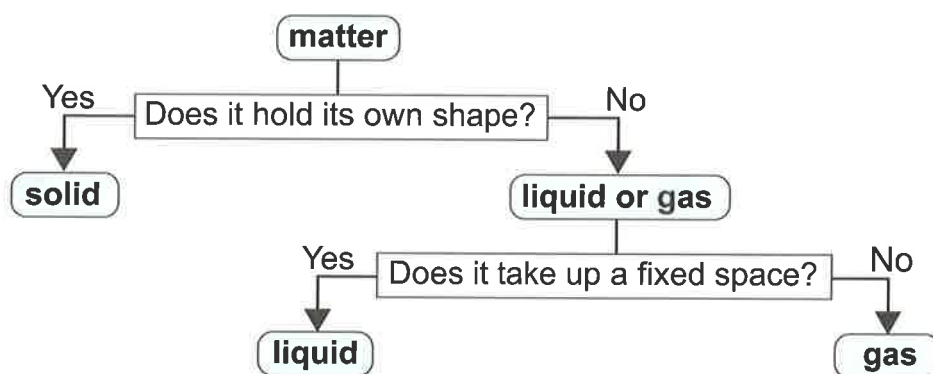


Which of the following shows the blocks in the correct order, from the lightest to the heaviest?

- (A) ● ▲ ■ (B) ■ ▲ ● (C) ■ ● ▲ (D) ▲ ■ ●

23. Everything around us is made up of matter. Matter can exist in three main states: solid, liquid or gas. For example, wood is a solid, water is a liquid, and air is a gas.

The flowchart helps to determine whether something is a solid, a liquid or a gas.



Karin has some ice cubes in a cup of drink.

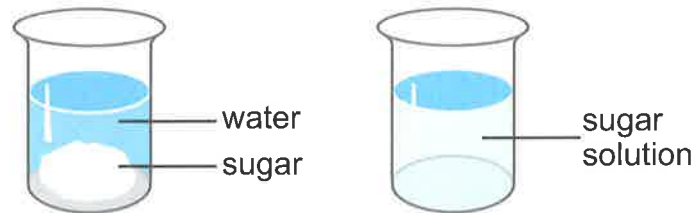


What states of matter are inside this cup?

- (A) liquid and gas only (B) liquid and solid only
(C) solid and gas only (D) solid, liquid and gas

For questions 24 and 25 use the information below.

When you add some sugar to a glass of water and stir for a while, the sugar crystals slowly disappear, leaving a clear liquid. We say the sugar dissolves in water, and the two together make a sugar solution.

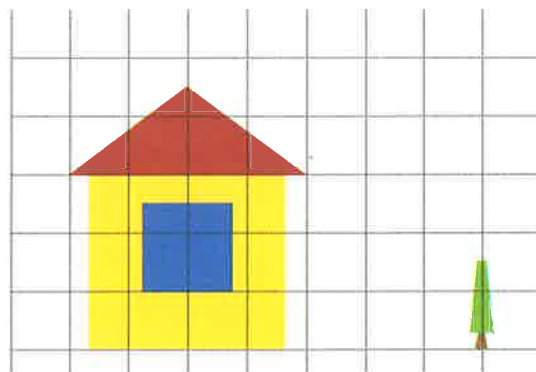


The table shows how much sugar can dissolve in 100 grams of water at different temperatures.

Temperature (°C)	10	20	40	60
Amount of sugar (grams)	180	200	240	280

24. How many grams of sugar can dissolve in 50 grams of water at 20 °C?
- (A) 100 (B) 150 (C) 200 (D) 250
25. Which statement is correct based on the table?
- (A) The more sugar dissolved in water, the higher the water temperature.
(B) The higher the water temperature, the more sugar can dissolve in the water.
(C) The higher the water temperature, the more water is in the solution.
(D) When the temperature doubles, the amount of sugar that can dissolve doubles.

26. Luke drew a house and a tree on square grid paper to compare their sizes.



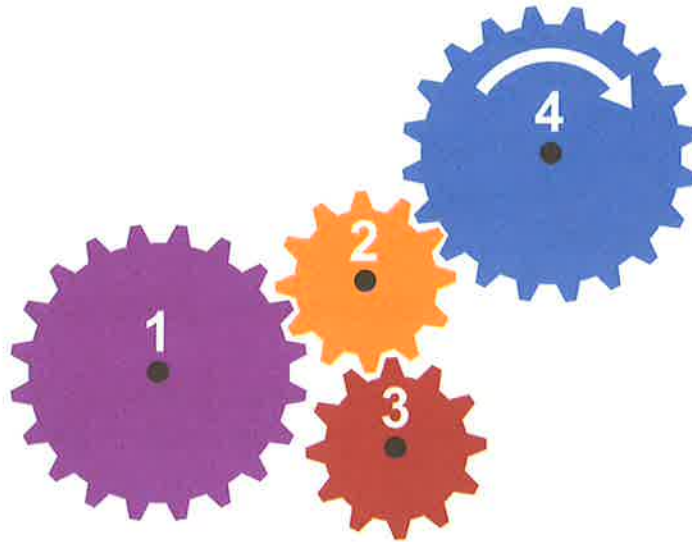
The tree is one and a half metres tall.

How many metres high is the top of the roof of the house?

- (A) 3 (B) 4 (C) $4\frac{1}{2}$ (D) $5\frac{1}{2}$

27. A gear is a wheel with teeth around its edge. When the teeth on the gears fit together, one gear can make another gear turn.

Alice put together a set of gears, as shown.



Which of these statements is correct?

- (A) Gears 1, 2 and 3 turn in the same direction.
- (B) Gear 1 turns in the same direction as gear 3.
- (C) Gear 3 turns in the opposite direction to gear 4.
- (D) Gear 4 turns in the opposite direction to gear 1.

28. Scientists made a new material with these features:

- waterproof
- cannot be bent
- does not burn
- can be heated up to 300 °C
- allows heat to pass through it.

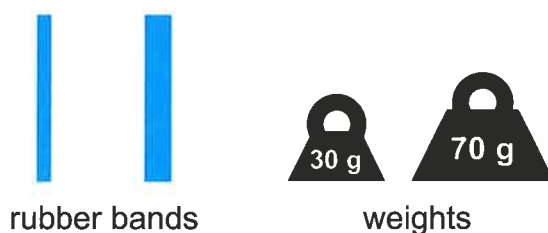
Which of these is the most suitable product that could be made using this new material?

- | | |
|--------------------|-----------------|
| (A) oven gloves | (B) garden hose |
| (C) drinking glass | (D) saucepan |

For questions 29 and 30 use the information below.

Mika's class is doing an experiment using some weights and rubber bands.

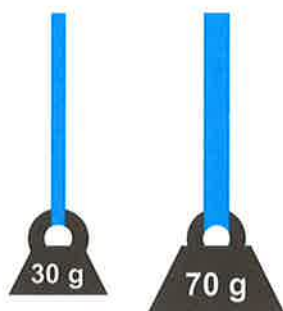
Mika has two weights and two rubber bands of the same length but of different widths. The weights can hang from the rubber bands.



29. Mika wants to find out whether the width of a rubber band affects how much it is stretched when a weight hangs from it.

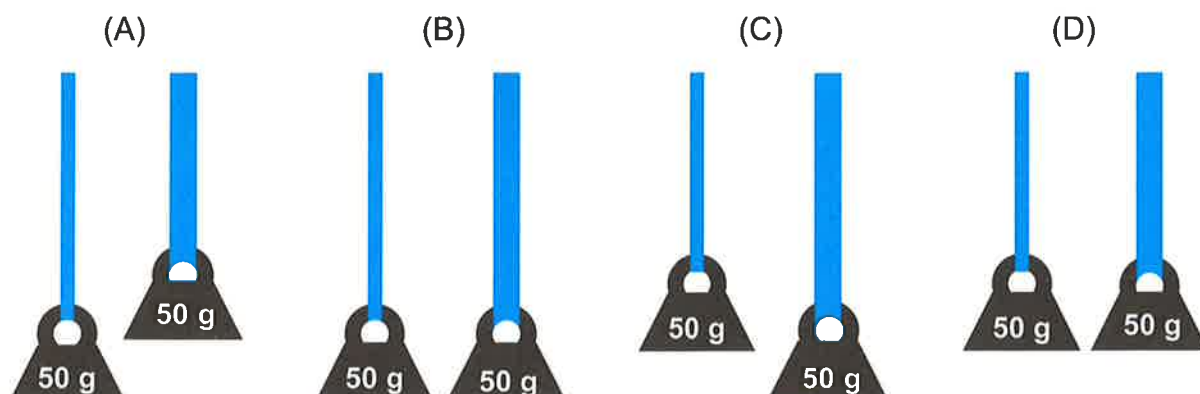
Which of the following items are required to investigate this?

- (A) both rubber bands and both weights
 - (B) both rubber bands and one weight
 - (C) one rubber band and both weights
 - (D) one rubber band and one weight
30. Mika's teacher hangs two weights from the end of two rubber bands and they stretch an equal amount, as shown.



She then replaces the weights with  .

What will the rubber bands look like now?



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Australia	Year 3
Brunei	Primary 3
Hong Kong	Primary 3
Indian Subcontinent¹	Class 3
Indonesia	Year 4
Malaysia	Standard 3
Middle East²	Class 3
New Zealand/Pacific³	Year 4
Singapore	Primary 2
Southern Africa⁴	Grade 3

¹ Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh.

² Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Egypt, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria and Jordan.

³ Pacific Region: Vanuatu, Papua New Guinea and Fiji.

⁴ Southern African Region: South Africa, Botswana, Lesotho, Swaziland, Zimbabwe and Namibia.



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AUSTRALIA

ANALYSIS BY QUESTION

The following table shows the questions you answered correctly ☒ and shades the questions that you answered incorrectly ☐. The questions are ranked from hardest at the top to easiest at the bottom.

	Question content	Area assessed	Question number	Correct answer
Difficult Questions	Identify the materials required for an investigation	Investigating	29	B
	Perform a calculation using data from a table	Interpreting	24	A
	Draw a conclusion using a flowchart	Predicting/Concluding	23	D
	Draw a conclusion based on information in a diagram	Investigating	13	B
	Identify the effect that temperature has on germination of different seeds	Interpreting	19	D
	Solve a problem using information provided	Reasoning/Problem solving	30	A
	Deduce the speed and direction of movement of a number of cogs	Reasoning/Problem solving	27	B
	Describe a household situation using scientific terminology	Investigating	14	D
	Identify the best design for an investigation	Investigating	18	D
	Rank objects on scales by mass from smallest to greatest	Reasoning/Problem solving	22	B
Easy Questions	Draw a conclusion based on information in a table	Predicting/Concluding	25	B
	Make a prediction using information in diagrams	Reasoning/Problem solving	20	A
	Use a scale diagram to find the height of a building	Observing/Measuring	26	C
	Predict the potential use of a product from its properties	Reasoning/Problem solving	28	D
	Deduce the positions of Earth, the Sun and the Moon from a description	Predicting/Concluding	21	C
	Estimate half of a length given a full length	Observing/Measuring	6	C
	Interpret data in a table	Interpreting	11	C
	Interpret data in a pie chart	Interpreting	4	B
	Find the difference in readings on two thermometers	Observing/Measuring	12	A
	Measure the difference between two volumes	Observing/Measuring	17	B
	Identify the information that will be needed in order to confirm a prediction	Investigating	15	D
	Identify the cause of an effect illustrated in a diagram	Reasoning/Problem solving	9	D
	Predict a letter using the flags in a semaphore system	Predicting/Concluding	8	C
	Recognise the position of a structure in a flower	Observing/Measuring	2	B
	Identify part of a process shown in a diagram	Predicting/Concluding	16	C
	Match a photo of a space object to its description	Observing/Measuring	3	A
	Find a value range from a table	Interpreting	5	C
	Interpret data in a table	Interpreting	10	C
	Identify the column graph that correctly represents data in a table	Interpreting	7	A
	Recognise the purpose of an action	Observing/Measuring	1	A